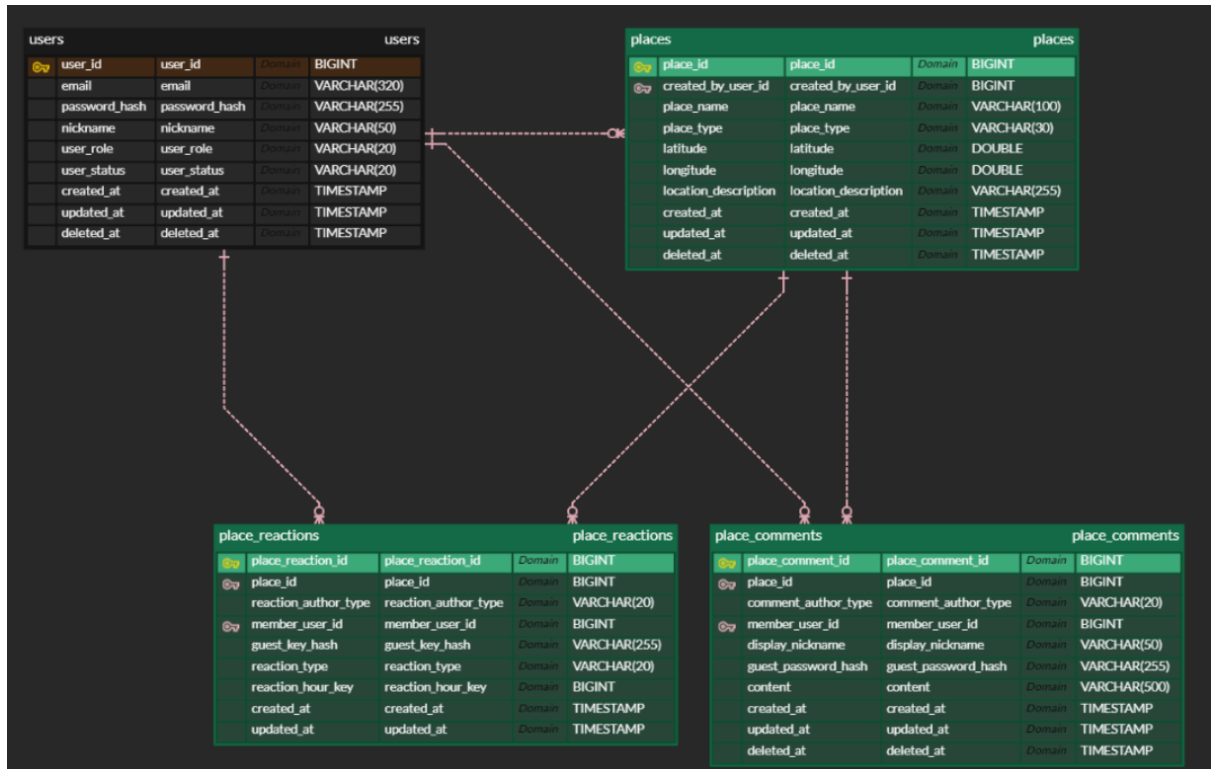




ERD

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users

```
CREATE TABLE users (  
  user_id BIGINT GENERATED BY DEFAULT AS IDENTITY,  
  
  email VARCHAR(320) NOT NULL,  
  password_hash VARCHAR(255) NOT NULL,  
  nickname VARCHAR(50) NOT NULL,  
  
  user_role VARCHAR(20) NOT NULL DEFAULT 'MEMBER',  
  user_status VARCHAR(20) NOT NULL DEFAULT 'ACTIVE',  
  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
);
```

```

updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),
deleted_at TIMESTAMPTZ NULL,

CONSTRAINT pk_users PRIMARY KEY (user_id),

CONSTRAINT chk_users_role
    CHECK (user_role IN ('MEMBER', 'ADMIN')),

CONSTRAINT chk_users_status
    CHECK (user_status IN ('ACTIVE', 'WITHDRAWN', 'BANN
ED')),

CONSTRAINT chk_users_email_not_blank
    CHECK (char_length(btrim(email)) > 0),

CONSTRAINT chk_users_password_hash_not_blank
    CHECK (char_length(btrim(password_hash)) > 0),

CONSTRAINT chk_users_nickname_not_blank
    CHECK (char_length(btrim(nickname)) > 0),

CONSTRAINT chk_users_updated_at
    CHECK (updated_at >= created_at),

CONSTRAINT chk_users_deleted_at
    CHECK (deleted_at IS NULL OR deleted_at >= created_
at)
);

```

places

```

CREATE TABLE places (
    place_id BIGINT GENERATED BY DEFAULT AS IDENTITY,

    created_by_user_id BIGINT NOT NULL,

    place_name VARCHAR(100) NOT NULL,
    place_type VARCHAR(30) NOT NULL,

```

```

latitude NUMERIC(10, 7) NOT NULL,
longitude NUMERIC(10, 7) NOT NULL,

-- PostGIS 거리 검색용 컬럼
-- ST_MakePoint는 longitude, latitude 순서
location GEOGRAPHY(Point, 4326) GENERATED ALWAYS AS (
    ST_SetSRID(
        ST_MakePoint(longitude::double precision, latitude::double precision),
        4326
    )::geography
) STORED,

location_description VARCHAR(255) NULL,

created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),
deleted_at TIMESTAMPTZ NULL,

CONSTRAINT pk_places PRIMARY KEY (place_id),

CONSTRAINT fk_places_created_by_user
    FOREIGN KEY (created_by_user_id)
    REFERENCES users (user_id)
    ON DELETE RESTRICT
    ON UPDATE CASCADE,

CONSTRAINT chk_places_type
    CHECK (
        place_type IN (
            'SMOKING_BOOTH',
            'SMOKING_AREA',
            'IMPLICIT_SMOKING_AREA',
            'NON_SMOKING_AREA'
        )
    ),

```

```

CONSTRAINT chk_places_latitude_range
    CHECK (latitude BETWEEN -90 AND 90),

CONSTRAINT chk_places_longitude_range
    CHECK (longitude BETWEEN -180 AND 180),

CONSTRAINT chk_places_name_not_blank
    CHECK (char_length(btrim(place_name)) > 0),

CONSTRAINT chk_places_location_description_not_blank
    CHECK (
        location_description IS NULL
        OR char_length(btrim(location_description)) > 0
    ),

CONSTRAINT chk_places_updated_at
    CHECK (updated_at >= created_at),

CONSTRAINT chk_places_deleted_at
    CHECK (deleted_at IS NULL OR deleted_at >= created_
at)
);

```

place_reactions

```

CREATE TABLE place_reactions (
    place_reaction_id BIGINT GENERATED BY DEFAULT AS IDENTI
TY,

    place_id BIGINT NOT NULL,

    reaction_author_type VARCHAR(20) NOT NULL,
    member_user_id BIGINT NULL,
    guest_key_hash VARCHAR(255) NULL,

    reaction_type VARCHAR(20) NOT NULL,

    reaction_hour_key BIGINT NOT NULL DEFAULT floor(extract

```

```

(epoch from now()) / 3600)::bigint,

created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),

CONSTRAINT pk_place_reactions PRIMARY KEY (place_reacti
on_id),

CONSTRAINT fk_place_reactions_place
    FOREIGN KEY (place_id)
    REFERENCES places (place_id)
    ON DELETE CASCADE
    ON UPDATE CASCADE,

CONSTRAINT fk_place_reactions_member_user
    FOREIGN KEY (member_user_id)
    REFERENCES users (user_id)
    ON DELETE RESTRICT
    ON UPDATE CASCADE,

CONSTRAINT chk_place_reactions_author_type
    CHECK (reaction_author_type IN ('MEMBER', 'GUES
T')),

CONSTRAINT chk_place_reactions_type
    CHECK (reaction_type IN ('LIKE', 'DISLIKE', 'CANCEL
ED')),

CONSTRAINT chk_place_reactions_author_rule
    CHECK (
        (
            reaction_author_type = 'MEMBER'
            AND member_user_id IS NOT NULL
            AND guest_key_hash IS NULL
        )
        OR
        (
            reaction_author_type = 'GUEST'

```

```

        AND member_user_id IS NULL
        AND guest_key_hash IS NOT NULL
    )
),

CONSTRAINT chk_place_reactions_guest_key_hash_not_blank
CHECK (
    guest_key_hash IS NULL
    OR char_length(btrim(guest_key_hash)) > 0
),

CONSTRAINT chk_place_reactions_hour_key
CHECK (reaction_hour_key >= 0),

CONSTRAINT chk_place_reactions_updated_at
CHECK (updated_at >= created_at)
);

```

place_comments

```

CREATE TABLE place_comments (
    place_comment_id BIGINT GENERATED BY DEFAULT AS IDENTITY,

    place_id BIGINT NOT NULL,

    comment_author_type VARCHAR(20) NOT NULL,
    member_user_id BIGINT NULL,

    display_nickname VARCHAR(50) NOT NULL,
    guest_password_hash VARCHAR(255) NULL,

    content VARCHAR(500) NOT NULL,

    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    deleted_at TIMESTAMPTZ NULL,

```

```

CONSTRAINT pk_place_comments PRIMARY KEY (place_comment
_id),

CONSTRAINT fk_place_comments_place
    FOREIGN KEY (place_id)
    REFERENCES places (place_id)
    ON DELETE CASCADE
    ON UPDATE CASCADE,

CONSTRAINT fk_place_comments_member_user
    FOREIGN KEY (member_user_id)
    REFERENCES users (user_id)
    ON DELETE RESTRICT
    ON UPDATE CASCADE,

CONSTRAINT chk_place_comments_author_type
    CHECK (comment_author_type IN ('MEMBER', 'GUEST')),

CONSTRAINT chk_place_comments_author_rule
    CHECK (
        (
            comment_author_type = 'MEMBER'
            AND member_user_id IS NOT NULL
            AND guest_password_hash IS NULL
        )
        OR
        (
            comment_author_type = 'GUEST'
            AND member_user_id IS NULL
            AND guest_password_hash IS NOT NULL
        )
    ),

CONSTRAINT chk_place_comments_display_nickname_not_blank
    CHECK (char_length(btrim(display_nickname)) > 0),

CONSTRAINT chk_place_comments_content_not_blank

```

```

        CHECK (char_length(btrim(content)) > 0),

CONSTRAINT chk_place_comments_guest_password_hash_not_b
lank
        CHECK (
            guest_password_hash IS NULL
            OR char_length(btrim(guest_password_hash)) > 0
        ),

CONSTRAINT chk_place_comments_updated_at
        CHECK (updated_at >= created_at),

CONSTRAINT chk_place_comments_deleted_at
        CHECK (deleted_at IS NULL OR deleted_at >= created_
at)
);

```

전체

```

-- PostgreSQL 기준
-- 좌표 거리 검색을 위해 PostGIS 사용 권장
CREATE EXTENSION IF NOT EXISTS postgis;

-- updated_at 자동 갱신용 함수
CREATE OR REPLACE FUNCTION set_updated_at()
RETURNS TRIGGER AS $$
BEGIN
    NEW.updated_at = now();
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TABLE users (
    user_id BIGINT GENERATED BY DEFAULT AS IDENTITY,

    email VARCHAR(320) NOT NULL,

```



```

password_hash VARCHAR(255) NOT NULL,
nickname VARCHAR(50) NOT NULL,

user_role VARCHAR(20) NOT NULL DEFAULT 'MEMBER',
user_status VARCHAR(20) NOT NULL DEFAULT 'ACTIVE',

created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),
deleted_at TIMESTAMPTZ NULL,

CONSTRAINT pk_users PRIMARY KEY (user_id),

CONSTRAINT chk_users_role
    CHECK (user_role IN ('MEMBER', 'ADMIN')),

CONSTRAINT chk_users_status
    CHECK (user_status IN ('ACTIVE', 'WITHDRAWN', 'BANN
ED')),

CONSTRAINT chk_users_email_not_blank
    CHECK (char_length(btrim(email)) > 0),

CONSTRAINT chk_users_password_hash_not_blank
    CHECK (char_length(btrim(password_hash)) > 0),

CONSTRAINT chk_users_nickname_not_blank
    CHECK (char_length(btrim(nickname)) > 0),

CONSTRAINT chk_users_updated_at
    CHECK (updated_at >= created_at),

CONSTRAINT chk_users_deleted_at
    CHECK (deleted_at IS NULL OR deleted_at >= created_
at)
);

CREATE TABLE places (

```

```

place_id BIGINT GENERATED BY DEFAULT AS IDENTITY,

created_by_user_id BIGINT NOT NULL,

place_name VARCHAR(100) NOT NULL,
place_type VARCHAR(30) NOT NULL,

latitude NUMERIC(10, 7) NOT NULL,
longitude NUMERIC(10, 7) NOT NULL,

-- PostGIS 거리 검색용 컬럼
-- ST_MakePoint는 longitude, latitude 순서다. 반대로 넣으면 지구 반대편으로 간다.
location GEOGRAPHY(Point, 4326) GENERATED ALWAYS AS (
    ST_SetSRID(
        ST_MakePoint(longitude::double precision, latitude::double precision),
        4326
    )::geography
) STORED,

location_description VARCHAR(255) NULL,

created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),
deleted_at TIMESTAMPTZ NULL,

CONSTRAINT pk_places PRIMARY KEY (place_id),

CONSTRAINT fk_places_created_by_user
    FOREIGN KEY (created_by_user_id)
    REFERENCES users (user_id)
    ON DELETE RESTRICT
    ON UPDATE CASCADE,

CONSTRAINT chk_places_type
    CHECK (
        place_type IN (

```

```

        'SMOKING_BOOTH',
        'SMOKING_AREA',
        'IMPLICIT_SMOKING_AREA',
        'NON_SMOKING_AREA'
    )
),

CONSTRAINT chk_places_latitude_range
    CHECK (latitude BETWEEN -90 AND 90),

CONSTRAINT chk_places_longitude_range
    CHECK (longitude BETWEEN -180 AND 180),

CONSTRAINT chk_places_name_not_blank
    CHECK (char_length(btrim(place_name)) > 0),

CONSTRAINT chk_places_location_description_not_blank
    CHECK (
        location_description IS NULL
        OR char_length(btrim(location_description)) > 0
    ),

CONSTRAINT chk_places_updated_at
    CHECK (updated_at >= created_at),

CONSTRAINT chk_places_deleted_at
    CHECK (deleted_at IS NULL OR deleted_at >= created_
at)
);

CREATE TABLE place_comments (
    place_comment_id BIGINT GENERATED BY DEFAULT AS IDENTIT
Y,

    place_id BIGINT NOT NULL,

    comment_author_type VARCHAR(20) NOT NULL,

```

```

member_user_id BIGINT NULL,

display_nickname VARCHAR(50) NOT NULL,
guest_password_hash VARCHAR(255) NULL,

content VARCHAR(500) NOT NULL,

created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),
deleted_at TIMESTAMPTZ NULL,

CONSTRAINT pk_place_comments PRIMARY KEY (place_comment
_id),

CONSTRAINT fk_place_comments_place
    FOREIGN KEY (place_id)
    REFERENCES places (place_id)
    ON DELETE CASCADE
    ON UPDATE CASCADE,

CONSTRAINT fk_place_comments_member_user
    FOREIGN KEY (member_user_id)
    REFERENCES users (user_id)
    ON DELETE RESTRICT
    ON UPDATE CASCADE,

CONSTRAINT chk_place_comments_author_type
    CHECK (comment_author_type IN ('MEMBER', 'GUEST')),

CONSTRAINT chk_place_comments_author_rule
    CHECK (
        (
            comment_author_type = 'MEMBER'
            AND member_user_id IS NOT NULL
            AND guest_password_hash IS NULL
        )
        OR
        (

```

```

        comment_author_type = 'GUEST'
        AND member_user_id IS NULL
        AND guest_password_hash IS NOT NULL
    )
),

CONSTRAINT chk_place_comments_display_nickname_not_blank
    CHECK (char_length(btrim(display_nickname)) > 0),

CONSTRAINT chk_place_comments_content_not_blank
    CHECK (char_length(btrim(content)) > 0),

CONSTRAINT chk_place_comments_guest_password_hash_not_blank
    CHECK (
        guest_password_hash IS NULL
        OR char_length(btrim(guest_password_hash)) > 0
    ),

CONSTRAINT chk_place_comments_updated_at
    CHECK (updated_at >= created_at),

CONSTRAINT chk_place_comments_deleted_at
    CHECK (deleted_at IS NULL OR deleted_at >= created_at);

CREATE TABLE place_reactions (
    place_reaction_id BIGINT GENERATED BY DEFAULT AS IDENTITY,

    place_id BIGINT NOT NULL,

    reaction_author_type VARCHAR(20) NOT NULL,
    member_user_id BIGINT NULL,
    guest_key_hash VARCHAR(255) NULL,

```

```

    reaction_type VARCHAR(20) NOT NULL,

    reaction_hour_key BIGINT NOT NULL DEFAULT floor(extract
(epoch from now()) / 3600)::bigint,

    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),

    CONSTRAINT pk_place_reactions PRIMARY KEY (place_reacti
on_id),

    CONSTRAINT fk_place_reactions_place
        FOREIGN KEY (place_id)
        REFERENCES places (place_id)
        ON DELETE CASCADE
        ON UPDATE CASCADE,

    CONSTRAINT fk_place_reactions_member_user
        FOREIGN KEY (member_user_id)
        REFERENCES users (user_id)
        ON DELETE RESTRICT
        ON UPDATE CASCADE,

    CONSTRAINT chk_place_reactions_author_type
        CHECK (reaction_author_type IN ('MEMBER', 'GUES
T')),

    CONSTRAINT chk_place_reactions_type
        CHECK (reaction_type IN ('LIKE', 'DISLIKE', 'CANCEL
ED')),

    CONSTRAINT chk_place_reactions_author_rule
        CHECK (
            (
                reaction_author_type = 'MEMBER'
                AND member_user_id IS NOT NULL
                AND guest_key_hash IS NULL

```

```

    )
    OR
    (
        reaction_author_type = 'GUEST'
        AND member_user_id IS NULL
        AND guest_key_hash IS NOT NULL
    )
),

CONSTRAINT chk_place_reactions_guest_key_hash_not_blank
CHECK (
    guest_key_hash IS NULL
    OR char_length(btrim(guest_key_hash)) > 0
),

CONSTRAINT chk_place_reactions_hour_key
CHECK (reaction_hour_key >= 0),

CONSTRAINT chk_place_reactions_updated_at
CHECK (updated_at >= created_at)
);

```